

A monovalent serotype refers to a single, specific variant of a microorganism (such as a virus or bacterium) characterized by its unique surface antigens, targeted by a single-strain (monovalent) diagnostic antiserum or vaccine.

The formulation step in vaccine production is the critical phase where the purified active antigen is combined with precise amounts of excipients, adjuvants, and stabilizers to create a safe, stable, and effective final mixture.

Critical Quality Attributes (CQAs) in vaccine manufacturing are measurable physical, chemical, biological, or microbiological properties that must stay within an approved range to guarantee product safety and effectiveness. Regulators like the European Medicines Agency (EMA) enforce these standards across the product lifecycle

Client Motivation

A Critical Quality Attribute (CQA) is a broader product property that ensures safety and efficacy, whereas a Critical Release Attribute (CRA) is a specific subset of a CQA tested on every batch to approve it for commercial distribution.

Critical Release Attribute (CRA)
Definition: A specific operational CQA that is chosen for routine batch testing and must meet pre-defined specification limits before a batch is released to the market.
Scope: Limited strictly to final batch release testing specifications.
Origin: Selected from established CQAs that are measurable right before packaging and shipment to confirm batch-to-batch consistency.

The client is a pharmaceutical manufacturer producing multivalent pneumococcal vaccines blended from monovalent serotype ‘bags’ at the formulation step. Antigenicity - the strength of the immune response a batch can elicit - is a critical release attribute. Antigenicity is a weighted property: each final batch’s antigenicity depends on which upstream bags were mixed and in what proportion.

Currently, the client can only measure antigenicity after blending and testing a batch. **The goal of this project is to build a predictive model that estimates a batch’s total antigenicity before mixing (formulation), using the known quality attributes (e.g., weighted parameter values) of the candidate upstream bags.** With such a model, batch planners could:

- Simulate different combinations of available bags before committing to a physical blend.
- Flag combinations likely to fall outside acceptable antigenicity specifications, reducing costly failed batches.
- Prioritise which bags to pair together to make the most efficient use of limited/aging inventory (e.g., bags nearing expiry).
- Provide a data-driven justification for batch composition decisions, supporting quality and regulatory review.

Identifying antigenicity specifications during vaccine manufacturing involves defining and testing critical quality attributes (CQAs) to ensure the antigen retains its correct structure, identity, and ability to bind target immune cells

In short, the client wants to move from reactive to predictive decision-making when composing batches from multiple pneumococcal variants by using historical batch genealogy and weighted parameter data as the training dataset.

The pneumococcal vaccine is also known as the pneumonia vaccine. It protects against pneumococcal disease.

Pneumococcal disease can lead to:
pneumonia
sepsis
meningitis

Process Overview

Pneumococcal conjugate vaccines (e.g., PCV13, PCV15, PCV20) confer protection against multiple serotypes of *Streptococcus pneumoniae*. Because each serotype requires its own polysaccharide purification and conjugation process, these vaccines are inherently multi-variant products, and this is reflected throughout the manufacturing process, from upstream drug substance production through fill/finish.

1. **Polysaccharide Production:** Each serotype is fermented, purified, and characterised independently, yielding a distinct drug substance lot (Master Batch Code, MBC) per serotype. A single final drug product batch is genealogically linked back to multiple upstream, serotype-specific batches, as illustrated in Figure 1.
2. **Conjugation:** Purified polysaccharides are chemically conjugated to a carrier protein (e.g., CRM197) to enhance immunogenicity, particularly in infant populations. Conjugation is run and released independently for each serotype, prior to any blending.
3. **Formulation:** Individual conjugated serotype ‘monovalent bulks’ are blended in defined ratios to form the multivalent drug product bulk. This is the step at which multiple upstream MBC lots converge into a single formulated product, shown in Figure 1 as the mapping from MBC lots to material numbers, and where traceability from the final batch back to each contributing serotype lot becomes critical for quality records.
4. **Fill/Finish:** The formulated bulk is filled into its final commercial presentation. A single formulated bulk may be filled into several distinct presentations (e.g., single-dose vial, multi-dose vial, or prefilled syringe), each assigned its own material number, and each material number is, in turn, produced across repeated batches over time. This two-stage fan-out - bulk-to-material and material-to-batch - is depicted in Figure 1.
5. **Quality Release & Genealogy:** Because each final batch is a composite of multiple upstream serotype lots, robust genealogy tracking (final batch → material number → upstream MBC lot) is essential for release testing, deviation investigations, and Annual Product Quality Reviews (APQR). An issue traced to a single serotype’s upstream lot can, via this genealogy, affect every downstream batch into which that lot was blended.

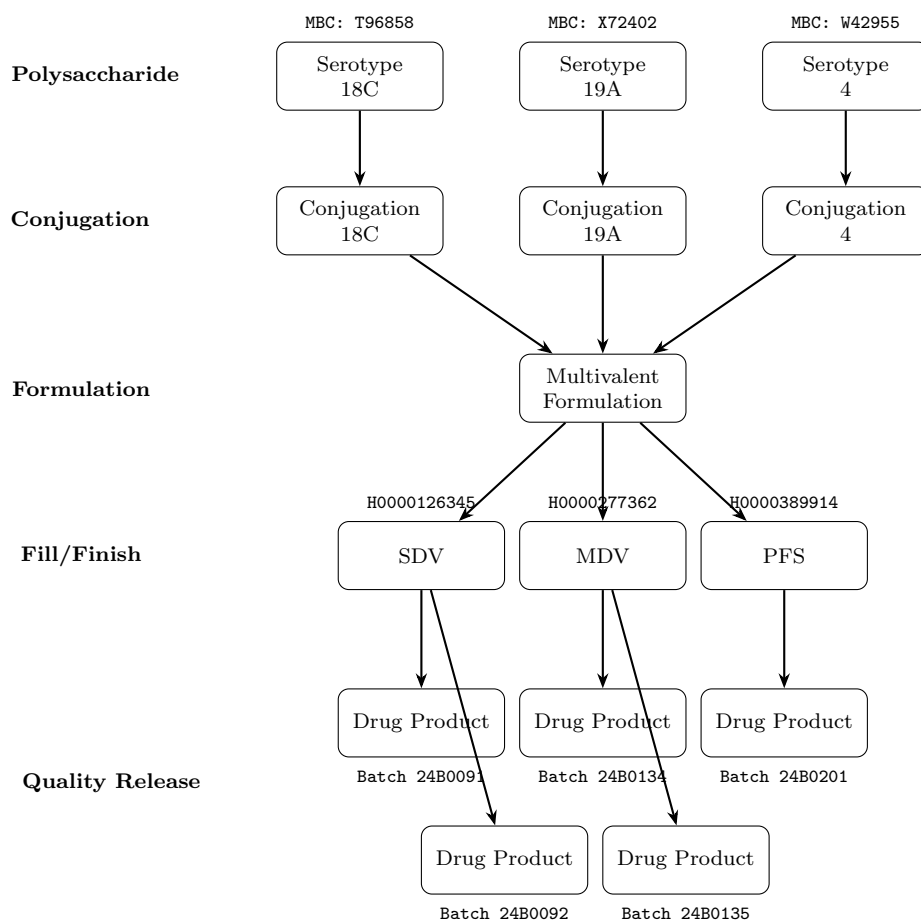


Figure 1: Identifier lineage from serotype-specific MBC lots through conjugation and blending, to fan-out across materials at Fill/Finish, and further fan-out to individual batch numbers at Release. A single blended bulk yields multiple materials, and each material is produced across multiple final batches. (MDV: Multi-dose vial; SDV: Single-dose vial; PFS: Pre-filled syringes).

Dataset Schema

Manufacturing data is available in stacked format. The service provider must unstack the data using 'Parameter' and 'Weighted Average Value' to pivot the table into a format suitable for modelling.

R or python

- **Batch Number**(VARCHAR): The drug product batch number.
- **Material Number**(VARCHAR): SDK code used to track inventory, link batch records, etc.
- **Manufacture Site**(VARCHAR): Site location. Conjugation may occur at one site, whilst packaging may occur elsewhere.
- **Product Name**(VARCHAR): The commercial drug product name.
- **Sub Product Name** (VARCHAR): Additional metadata about the product name; in this case, that the final output is a drug product.
- **Parameter**(VARCHAR): The names of measurements taken during the manufacturing process of multivalent vaccines.
- **Weighted Average Value** (FLOAT): The numerical value recorded for the Parameter name.
- **MBC Batch Number**(VARCHAR): The drug substance MBC lot code associated with monovalent serotype bags.
- **Date of Manufacture**(TIMESTAMP): Date of measurement recorded for Parameter & Weighted Average Value.

Please note that the client has performed Genealogy linkage analysis and calculated the weighted average value for measurements taken during the manufacturing process. Batch genealogy is outside of the scope of this project.

VARCHAR: a variable-length character data type used in databases to store strings of letters, numbers, and symbols. variable character

FLOAT: a data type used in programming and databases to represent numbers with fractional parts or decimal points